n-nim Vector

The n-dim vector is defined as follow:

$$X \in \mathbb{R}^n$$

Note that

$$\mathbb{R}^n = \left\{ (x_1, \cdots, x_n) | x_i \in \mathbb{R} \right\}$$

Matrix

The $n \times m$ matrix is defined as follow:

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = [a_{ij}]_{m \times n}$$

each a_{ij} is a real number.

Determinant

Given a matrix $M = [m_{ij}]_{2\times 2}$, the determinant of the matrix is

$$\det M = \begin{vmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{vmatrix} = m_{11}m_{22} - m_{12}m_{21}$$

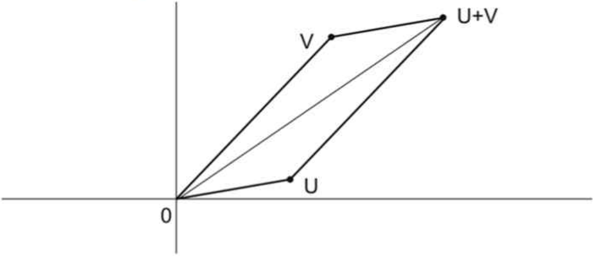
Given a matrix $M = [m_{ij}]_{3\times 3}$, the determinant of the matrix is

$$\det M = m_{11} \cdot \begin{vmatrix} m_{22} & m_{23} \\ m_{32} & m_{33} \end{vmatrix} - m_{12} \cdot \begin{vmatrix} m_{21} & m_{23} \\ m_{31} & m_{33} \end{vmatrix} + m_{13} \cdot \begin{vmatrix} m_{21} & m_{22} \\ m_{31} & m_{32} \end{vmatrix}$$

Sum of Vector

For two n-dim vector U and V,
$$U + V = (u_1 + v_1, \cdots, u_n + v_n)$$

The geometric meaning of constant multiplication is as follow:

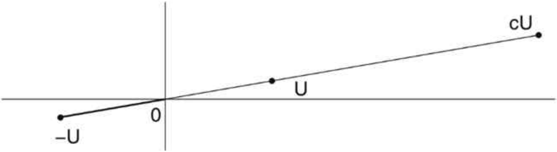


Constant Multiplication of Vector

For n-dim vector U and constant c,

$$cU = (cu_1, \cdots, cu_n)$$

The geometric meaning of constant multiplication is as follow:



If $c = 1$, U is U itself. If $c < 0$, the direction of U is reversed.

Linear Combination

The linear combination of n-dim vector X_i is as follow:

$$\sum_{i=1}^{n} c_i X_i \text{ where } c_i \in \mathbb{R}$$

Linearly Independent

The n-dim vector $X_1, \cdots, X_k$ is linearly independent if and only if

$$0 = \sum_{i=1}^{k} c_i X_i \Leftrightarrow c_i = 0$$

K-dim Plane

Let $U_1, \cdots, U_k$ be linearly independent vectors in $\mathbb{R}^n$. Then, the set of the following is called a k-dim plane

$$U + t_1 U_1 + \cdots + t_k U_k$$

When $k = n - 1$, we call it hyperplane.

When $U = 0$, we call the k-dim plane the span of $U_1, \cdots, U_k$

Linear Function

A function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called linear if and only if

$$① \; F(cV) = cF(V)$$

$$② \; F(V+W) = F(V) + F(W)$$

Every such linear function can be expressed by

$$F(V) = MV$$

where M is some $k \times n$ matrix

Norm

For vector $V = (v_1, \cdots, v_n)$, the norm is defined as follow:

$$\|V\| = \sqrt{\sum_{i=1}^{n} v_i^2}$$

For matrix $M = [m_{ij}]_{n \times m}$,

$$\|M\| = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{m} m_{ij}^2}$$

Dot Product

For vector V,U, the dot product is defined as follow:

$$V \cdot U = \sum_{i=1}^{n} (v_i \cdot u_i)$$

Cross Product

For 3-dim vector V,U, the cross product is defined as follow:

$$V \times U = \det \begin{vmatrix} i & j & k \\ v_1 & v_2 & v_3 \\ u_1 & u_2 & u_3 \end{vmatrix}$$

Triangle Inequality

For vector V,U, the following is true:

$$\|\mathrm{V}\cdot\mathrm{U}\| \leq \|\mathrm{V}\| + \|\mathrm{U}\|$$

Orthogonal Vector

For vector V,U, if

$$V \cdot U = 0$$

then those two vectors are orthogonal. We simply write it $V \perp U$

Orthonormal Set

A list of vectors $V_1, \cdots, V_m \in \mathbb{R}^n$ is called an orthonormal set when two properties hold:

① $V_j \perp V_k$ for all $j \neq k$

② $\|V_j\| = 1$ for all j

Cauchy Swartz Inequality

For vector V,U, the following is true:

$$|\mathrm{U} \cdot \mathrm{V}| \leq \|\mathrm{U}\| \cdot \|\mathrm{V}\|$$

Angle of Two Vectors

For vector V,U, the angle θ is defined as follow:

$$\mathrm{U} \cdot \mathrm{V} = \|\mathrm{U}\| \cdot \|\mathrm{V}\| \cos\theta$$

If $\theta = \dfrac{\pi}{2}$, then $\mathrm{U} \cdot \mathrm{V} = 0$. So the two vectors are orthogonal.

Signed Volume

Given three vector V_1, V_2, V_3, the signed volume created by those three vectors is

$$\frac{1}{3!}\det(V_1, V_2, V_3)$$